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Non-Specificity of Analogical Inference in High-Energy Astrophysics: A TNSA Case Study of GRB 241025A

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Abstract

This paper applies the Test of Non-Specificity of Analogies (TNSA) to GRB 241025A, a well-observed gamma-ray burst that has been interpreted within at least two distinct theoretical frameworks: a standard synchrotron afterglow model emphasizing radio colour evolution (Giarratana et al., 2026) and a Binary-driven HyperNova (BdHN I) classification (Ruffini et al.). We develop a structured TNSA analysis showing how the same multi-wavelength dataset supports multiple, internally coherent narratives. We examine the constraining power of radio spectral index evolution, quantify parameter degeneracies in the synchrotron model, and identify which additional observations would most increase model specificity. We conclude with a methodological checklist for applying TNSA to astrophysical events and propose concrete observational tests that would reduce non-specificity for GRB 241025A and similar bursts.

Keywords: Gamma-ray bursts; GRB 241025A; Radio afterglow; Spectral index evolution; Synchrotron afterglow; Binary-driven HyperNova; Model degeneracy; Test of Non-Specificity of Analogies (TNSA).

1 Introduction

Analogical inference is ubiquitous in scientific practice: researchers map new observations onto established theoretical structures by analogy with prior cases. In high-energy astrophysics, analogies underpin classification (e.g., long vs short GRBs, collapsar vs binary progenitors) and model selection (e.g., external shock synchrotron vs multi-component central engine scenarios). The Test of Non-Specificity of Analogies (TNSA) is a protocol designed to evaluate the evidential value of such analogical inferences by asking whether the same observational pattern can be reproduced by multiple, structurally distinct models. When multiple models can account for the same data, the analogical inference is non-specific and its evidential weight is reduced.

GRB 241025A is an instructive case. Detected by Fermi GBM and Swift, and followed up across X-ray, optical/NIR and radio bands, it has been analyzed in at least two published frameworks. Giarratana et al. (2026) present a multi-wavelength afterglow analysis that emphasizes the time evolution of the radio spectral index (“radio colour evolution”) and argues that a standard synchrotron external shock model can reasonably describe the data. Ruffini et al. propose a BdHN I interpretation that situates the burst within a binary progenitor and

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hypernova narrative. Both claims are internally coherent and supported by subsets of the data. This paper uses GRB 241025A to illustrate how TNSA can be applied to reveal non-specificity, to quantify which constraints are decisive, and to propose targeted observations that would increase specificity.

2 Background: TNSA and analogical inference

TNSA operationalizes the idea that analogical inferences have evidential value only to the extent that they are specific: a mapping from data to theory is informative when it is unlikely to be reproduced by alternative, independently motivated models. The protocol proceeds in three steps:

1. **Identify the target pattern** — the set of observational features to be explained (here: radio spectral index evolution $\alpha(t)$, optical/NIR light curves, X-ray decay indices, SED shape).

The first step requires a careful and explicit specification of the empirical pattern that the competing models must reproduce. For GRB 241025A we treat the target pattern as a composite object: the time series of radio spectral indices measured at multiple epochs, the contemporaneous optical and near-infrared light curves with their decay slopes and possible bumps, the X-ray temporal and spectral behaviour, and the broadband spectral energy distributions at selected epochs. Each of these elements carries different constraining power and different susceptibilities to observational systematics; TNSA requires that they be listed and weighted explicitly before model comparison proceeds.

2. **Enumerate competing models** — plausible, independently motivated theoretical frameworks that could account for the pattern (here: Synchrotron External Shock; BdHN I).

The second step is to assemble a set of candidate explanations that are motivated independently of the specific dataset. For GRB 241025A we consider at minimum the standard external shock synchrotron model, which is grounded in decades of afterglow phenomenology, and the BdHN I framework, which is motivated by a distinct progenitor narrative and a different set of prior case studies. TNSA emphasizes that competing models should be described in sufficient detail to allow concrete predictions to be derived, and that their independent motivations be documented so that the comparison is not merely ad hoc.

3. **Test non-specificity** — determine whether each model can reproduce the target pattern under plausible parameter choices; if multiple models can, quantify the degree of degeneracy and identify observations that would discriminate between them.

The third step operationalizes non-specificity: for each model we map the region of parameter space that yields acceptable fits to the target pattern, quantify overlaps in observational consequences across models, and identify which additional measurements would most effectively shrink the acceptable regions. This step requires both statistical tools (posterior sampling, profile likelihoods) and conceptual clarity about which aspects of the data are decisive versus which are flexible.

TNSA emphasizes explicitness: analogies used to justify a model must be stated, the mapping from data to model must be made explicit, and the space of alternative mappings must be explored.

3 Data summary for GRB 241025A

We summarize the observational record relevant to the afterglow interpretation. (For full data tables and references see Giarratana et al., 2026; GCN circulars 37860 and follow-ups.)

- **Prompt gamma-ray:** The burst was detected by Fermi GBM at 2024-10-25T01:36:26.9 UT; GCN notices report the burst duration (T90), fluence, and basic spectral parameters. These prompt properties set the energetic scale for afterglow modelling and provide priors on the initial kinetic energy and possible central engine activity that may influence late-time behaviour.
- **X-ray:** Swift-XRT began observations roughly 126 seconds after trigger and recorded a light curve that can be modelled by broken power laws with an early steep decay followed by shallower segments. The X-ray spectral indices and temporal breaks provide constraints on the cooling regime and on the presence of energy injection or late central engine activity, and they serve as an anchor for broadband SED fits.
- **Optical/NIR:** Swift-UVOT and several ground-based facilities obtained multi-band photometry spanning from minutes to days after the burst. The optical/NIR data constrain the location of the characteristic synchrotron frequency and can reveal signatures of a supernova component or host extinction; they are therefore crucial to distinguish between a simple external shock and more complex progenitor scenarios.
- **Radio:** A campaign of multi-frequency radio observations, covering several GHz bands over days to weeks, measured the instantaneous spectral index $\alpha(t)$ at multiple epochs. The radio data show a clear evolution from a flatter or positive index at early times—consistent with partial self-absorption or a low characteristic frequency—to a negative index at later times, consistent with optically thin synchrotron emission; the timing and slope of this transition are central to the TNSA analysis.

The key empirical feature for this study is the radio spectral index evolution: early epochs show a flatter or positive index consistent with partial self-absorption, while later epochs show a negative index consistent with optically thin synchrotron emission. The timing and slope changes are the primary constraints used by Giarratana et al. in their synchrotron interpretation.

4 Model A: Synchrotron external shock (Giarratana et al.)

4.1 Model description

The standard external shock synchrotron model assumes a relativistic blast wave interacting with an external medium. Electrons accelerated at the shock follow a power-law distribution $N(\gamma) \propto \gamma^{-p}$ above a minimum Lorentz factor γ_m . The synchrotron spectrum is characterized by three break frequencies: self-absorption ν_a , characteristic ν_m , and cooling ν_c . The flux density scales as $F_\nu \propto \nu^\alpha$ with α determined by the spectral regime.

Key model parameters:

$$\{E_k, n, \epsilon_e, \epsilon_B, p, \theta_j\},$$

where E_k is kinetic energy, n ambient density, ϵ_e and ϵ_B microphysical fractions, p electron index, and θ_j jet opening angle.

4.2 Fitting strategy and results (summary)

Giarratana et al. fit multi-wavelength SEDs and light curves, using MCMC sampling to explore parameter space. Their principal findings are that a coherent parameter set exists which reproduces the observed radio spectral index evolution together with the optical and X-ray light curves. The authors show that the radio colour evolution can be interpreted as the passage of a characteristic synchrotron break (either ν_m or ν_a depending on epoch) through the radio band, and that the inferred microphysical parameters fall within ranges commonly reported for GRB afterglows, albeit with substantial posterior uncertainties reflecting degeneracies and observational noise.

4.3 Parameter degeneracies

We analyze the degeneracies that remain even after multi-wavelength fitting. First, the kinetic energy E_k and the ambient density n trade off: increasing the energy while decreasing the density can preserve the observed peak flux and break frequencies, producing elongated posterior contours in the (E_k, n) plane. Second, the microphysical fractions ϵ_e and ϵ_B enter the expressions for characteristic frequencies and peak flux in coupled combinations, so that different pairs of values can yield similar SED shapes; this degeneracy is particularly acute when only limited spectral coverage is available. Third, the electron index p interacts with the assignment of spectral regimes: a small change in p can be compensated by reassigning which spectral segment corresponds to a given observational band, thereby preserving the fit quality while altering physical interpretation. These degeneracies persist despite the added constraint of radio colour evolution because the same temporal sequence of spectral slopes can be produced by different sequences of break passages or by modest time dependence in microphysical parameters.

5 Model B: BdHN I interpretation (Ruffini et al.)

5.1 Model description

The BdHN I framework posits a binary progenitor system in which a massive star collapses in the presence of a compact companion, producing a complex sequence of interactions that generate prompt gamma rays, a hypernova-like explosion, and multi-component late emission. The model emphasizes a narrative mapping from observed temporal and spectral features to stages of the binary interaction and ejecta dynamics, and it often invokes dense circumbinary environments and energy transfer between components to explain late-time behaviour.

5.2 How BdHN accounts for the data

Ruffini et al. argue that the prompt gamma-ray temporal structure, the energetics, and certain late optical features of GRB 241025A are consistent with a BdHN I classification. Within this narrative, radio emission can arise from the interaction of multiple ejecta components with a dense and structured circumbinary medium; the observed radio colour evolution can be reproduced by a combination of emitting regions with different characteristic frequencies and by time-dependent energy injection from the central engine or from collisions between ejecta shells. The BdHN interpretation therefore accommodates the multi-wavelength dataset by embedding the afterglow within a richer progenitor story that links prompt and late emission through physically motivated, if more complex, mechanisms.

5.3 Model flexibility and degrees of freedom

BdHN models typically allow multiple emission components, structured density profiles, and episodes of energy injection, which increases their flexibility to fit diverse observational patterns.

This flexibility is a double-edged sword: it enables the model to reproduce complex datasets, but it also increases the number of free parameters and reduces falsifiability unless the model makes specific, testable predictions (for example, the presence of supernova spectral lines at particular epochs, a characteristic late-time radio morphology, or host galaxy signatures tied to binary progenitors). TNSA therefore treats BdHN as a plausible alternative but emphasizes the need for discriminating observations to avoid overfitting.

6 Comparative TNSA analysis

6.1 Step 1: Target pattern

We formalize the target pattern P as:

$$P = \{\alpha(t_i) \text{ at radio epochs } t_i, F_\nu^{\text{opt}}(t), F_\nu^{\text{X}}(t), \text{ SED shape at selected epochs}\}.$$

This formalization makes explicit which empirical features must be matched and allows us to treat observational uncertainties and epoch matching systematically when comparing models.

6.2 Step 2: Model reproduction

We ask whether Model A and Model B can reproduce P under plausible parameterizations. For Model A, Giarratana et al. provide explicit fits and posterior samples demonstrating that a standard external shock synchrotron model can reproduce the radio spectral index evolution together with optical and X-ray light curves, using parameter values that are physically plausible and consistent with the broader GRB afterglow population. For Model B, Ruffini et al. present a BdHN narrative and parameter choices that can also accommodate the same multi-wavelength pattern, typically by invoking additional emitting components or environmental structure; their account is less compact but remains internally coherent.

6.3 Step 3: Degree of non-specificity

We quantify non-specificity qualitatively and propose a simple metric. Let S_M denote the region of model parameter space that reproduces P within observational uncertainties for model M . Non-specificity is high when both S_A and S_B are substantial and when their observational consequences overlap in ways that cannot be discriminated by the current dataset. While precise volumes require detailed posterior sampling and model evidence calculations, the existence of two independent, peer-reviewed fits indicates that both S_A and S_B are non-empty and that the mapping from P to a unique theoretical narrative is not injective. TNSA therefore classifies the current evidential situation as one of non-specificity and recommends targeted observations to reduce it.

7 Which observations would increase specificity?

TNSA is most useful when it suggests targeted observations that reduce non-specificity. For GRB 241025A we identify the following high-value data and explain why each would be discriminating.

1. **Late-time radio imaging and spectral monitoring (VLBI).** High-resolution radio imaging that spatially resolves the source or measures its expansion rate would directly test whether the radio emission arises from a single, expanding external shock or from multiple, spatially distinct ejecta components. VLBI observations can reveal source size evolution, proper motion, or structural asymmetries that are difficult to reconcile with a single-component synchrotron model but are natural in multi-component BdHN scenarios;

combined with spectral monitoring, such imaging would strongly constrain the geometry and emission sites.

2. **Polarization measurements (radio and optical).** Polarization encodes information about magnetic field geometry and the coherence scale of the emitting region. A standard external shock with shock-generated, tangled fields predicts low polarization or specific temporal evolution, whereas a scenario with ordered fields or multiple components can produce higher and more variable polarization signatures. Simultaneous polarization monitoring across bands would therefore discriminate between competing magnetic field and emission-region hypotheses.
3. **Deep late-time optical spectroscopy of the host and potential supernova features.** Detecting supernova spectral lines or a late-time bump in the optical spectrum would provide direct evidence for a hypernova component associated with a BdHN progenitor. Conversely, the absence of such features to deep limits would disfavour BdHN interpretations that predict a luminous supernova, thereby increasing the relative specificity of the synchrotron external shock model.
4. **Dense early radio cadence to characterize scintillation.** Interstellar scintillation can mimic or obscure intrinsic spectral evolution at early times, particularly at low radio frequencies. A dense cadence of radio observations in the first days post-burst would allow separation of scintillation-induced variability from genuine spectral index changes, thereby clarifying whether the observed radio colour evolution is intrinsic to the source or contaminated by propagation effects.
5. **Broadband SEDs at matched epochs (radio–X).** Simultaneous or near-simultaneous broadband spectral energy distributions reduce degeneracy in break assignment by fixing the relative positions of ν_a , ν_m , and ν_c at specific times. Matched-epoch SEDs therefore tighten constraints on microphysical parameters and reduce the volume of parameter space consistent with the data, improving the discriminating power between single-component synchrotron and multi-component BdHN models.

We rank these by expected discriminating power: VLBI imaging and polarization are highest because they probe geometry and magnetic structure directly, followed by late-time spectroscopy which tests progenitor predictions, and dense radio cadence and matched SEDs which reduce observational ambiguities and parameter degeneracies.

8 Practical TNSA checklist for astrophysical events

From this case study we extract a practical checklist for applying TNSA to transient astrophysical events. Each item below is accompanied by a short operational explanation to guide implementation.

1. **Explicitly state the analogies used to map data to theory.** For each model, document the analogical steps that map observed features to theoretical constructs (for example: “radio spectral index evolution $\rightarrow$ passage of ν_m ”). Making these mappings explicit prevents hidden assumptions and clarifies where analogical inference is doing the explanatory work.
2. **Enumerate alternative models and their independent motivations.** List competing frameworks that are motivated by prior theory or independent datasets, and summarize why each is plausible a priori. This prevents circular reasoning where a model is only considered because it fits the current data.

3. **Quantify parameter degeneracies within each model (posterior sampling, profile likelihoods).** Use MCMC or nested sampling to map posterior distributions and identify elongated or multi-modal regions; report profile likelihoods for key parameters to show which combinations are constrained and which remain degenerate.
4. **Identify discriminating observations and prioritize them by feasibility and expected information gain.** For each candidate observation, estimate the expected reduction in model-space volume (or an information-theoretic proxy) and weigh this against observational cost and feasibility to produce a prioritized follow-up plan.
5. **Report model-space volumes (or proxies) that reproduce the data to assess non-specificity.** Where possible, provide quantitative measures of the size of parameter regions consistent with the data (e.g., credible volume, Bayes factors, or likelihood-ratio thresholds) so that non-specificity can be assessed numerically rather than qualitatively.
6. **Publish reproducible fits and data products to enable independent TNSA replication.** Release posterior samples, SED fits, and reduced data in machine-readable form so that other researchers can reproduce the analysis, test alternative models, and apply TNSA independently.

9 Discussion

GRB 241025A exemplifies how a single dataset can support multiple, coherent scientific narratives. The radio colour evolution is a valuable constraint that strengthens internal consistency checks for synchrotron fits, but it does not by itself uniquely select a progenitor or emission scenario. BdHN models and synchrotron external shock models differ in explanatory scope and parsimony: BdHN offers a broader progenitor narrative while synchrotron models offer a compact, mechanistic account of afterglow emission. TNSA does not aim to declare winners; rather, it clarifies where evidential weight is thin and where targeted observations can most effectively increase specificity.

10 Conclusions

Applying TNSA to GRB 241025A reveals substantial non-specificity: both a standard synchrotron afterglow model and a BdHN I narrative can reasonably account for the observed multi-wavelength pattern, including radio colour evolution. We identify concrete observations (VLBI imaging, polarization, late-time spectroscopy) that would most reduce non-specificity. More broadly, TNSA provides a structured protocol for assessing the evidential value of analogical inferences in astrophysics and for prioritizing follow-up observations that increase model discriminability.

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